STATISTICAL MODELLING AND DESIGN OF EXPERIMENTS

Assignment 1

Joel González Jiménez

Jose Arias Perez

Quentin Meneghini

14th April 2026

Facultat d'Informàtica de Barcelona

Contents

Introduction **1**

1 Questions 1 **2**

1.1 Q1-a 2

1.2 Q1-b 2

1.3 Q1-c 4

1.4 Q1-d 5

1.5 Q1-e 5

1.6 Q1-f 6

1.7 Q1-g 6

1.8 Q1-h 7

1.9 Q1-i 7

1.10 Q1-j 8

1.11 Q1-k 8

2 Questions 2 **10**

2.1 Q2-a 10

2.2 Q2-b 10

2.3 Q2-c 13

2.4 Q2-d 13

2.5 Q2-e 13

3 Questions 3 **16**

3.1 Q3-a 16

3.2 Q3-b 18

3.3 Q3-c 20

3.4 Q3-d 22

4 Questions 4 **23**

4.1 Q4-a 23

4.2 Q4-b 27

4.3 Q4-c 29

Conclusion **31**

A Appendix section **32**

Introduction

This report presents a statistical analysis of an airplane dataset. We analyzed numerical and categorical variables to predict airplane prices and understand their relationships.

In the first section, we imported the data, summarized variables like Price and Fuel Consumption, and performed t-tests and chi-square tests. In the second section, we analyzed the effect of the airplane Model, Sales Region, and Production Year on Price using one-way and two-way ANOVA.

In the third section, we fitted Simple and Multiple Linear Regression (MLR) models to predict Price. We tested regression assumptions, applied transformations, and validated the final model using an 80/20 train-test split. In the final section, we applied Principal Component Analysis (PCA) to reduce dimensionality. We then fitted a Principal Component Regression (PCR) model and compared its performance against the MLR model.

1 Questions 1

1.1 Import data set to R assigning the type of each variable correctly

We casted to factor both the string types [”Model”, ”SalesRegion”] and the very repetitive columns [”NumberofEngines”, ”EngineType”]. We considered casting also ”ProductionYear”, but in later calculation as the median was required, so we decided to leave as is (numerical). The code is shown below.

```
air_data <- read.csv("./dataset.csv", sep=",", stringsAsFactors=TRUE)


air_data <- air_data %>%
  rename(
    FC = FuelConsumption.L.h.,
    HM = HourlyMaintenance...,
    Price = Price...
  )


to_factors <- c("Model", "NumberofEngines", "EngineType", "SalesRegion")


for (col in to_factors) {
  air_data[[col]] <- as.factor(air_data[[col]])
}
```

1.2 Summarize the variables Price, Fuel Consumption first for the all data and then for the groups of Engine Type

We reported the result of the summaries of both Price and Fuel Consuption in the tables 1 and 2, while an histogram is in figure 1

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
145815	14096814	83921914	198833650	384323881	978213229

Table 1: Price

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.00	5.95	9.82	12.08	13.47	49.97

Table 2: Fuel Consumption (FC)

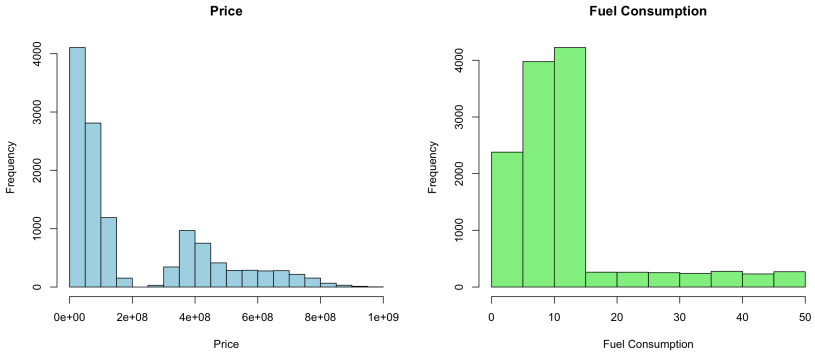

Figure 1: Histograms

Here are the same summaries but based on the Engine Type (Table 3 and Figure 2).

EngineType	Price_Mean	Price_SD	Price_Median	Price_Min	Price_Max	FC_Mean	FC_SD	FC_Median	FC_Min	FC_Max	n
Piston	279405.	77369.	251474.	145815.	561279.	30.1	11.6	29.9	10.0	50.0	2039
Turbofan	237995200.	231292861.	108876395.	8150222.	978213229.	8.52	3.75	8.53	2.0	15.0	10338

Table 3: Summary by Model

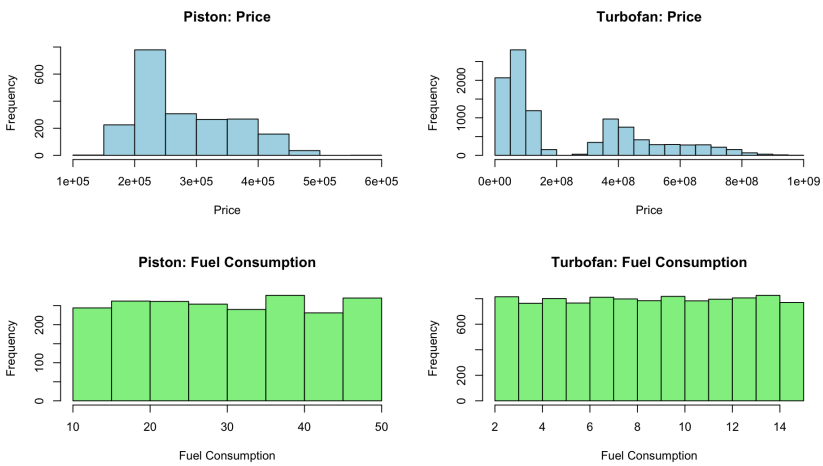

Figure 2: Summary distributions by Model

1.3 Test whether Fuel Consumption is affected from the Engine Type of the plane. Check the assumptions and visualize the relationship between these two characteristics

```
        Welch Two Sample t-test

data:  FC by EngineType
t = 83.347, df = 2123.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Piston and group Turbofan is not equal to 0
95 percent confidence interval:
 21.09784 22.11459
sample estimates:
  mean in group Piston mean in group Turbofan
            30.122398               8.516185
```

Figure 3: T-test results for Fuel Consumption by Engine Type

We performed a t-test and obtained a p-value, the results can be found in the figure 3. Since the p-value is < 0.05, we reject the null hypothesis and conclude that there is a significant difference in mean fuel consumption between the engine types. Furthermore we used a visual interpretation to be sure about the result as the number of observation made the test not very significant (figures 4 and 5).

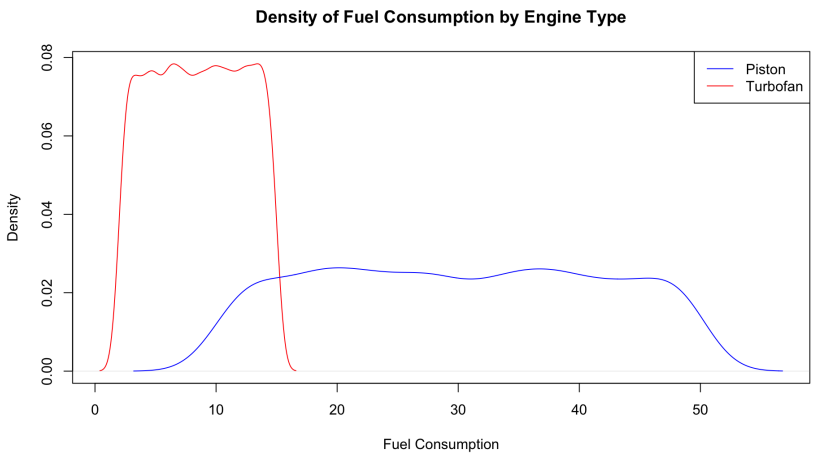

Figure 4: Comparing distributions

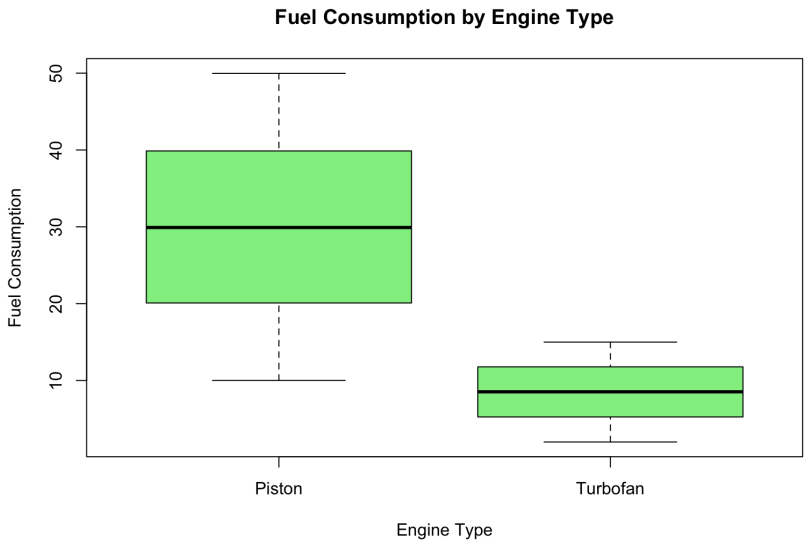

Figure 5: Boxplot is best for comparing a continuous variable across categories

Test assumptions

- We checked for normality (Shapiro-Wilk) and homogeneity of variances (Levene's test). Because the sample size is large (N > 5000), the normality test is overly sensitive, but the large sample size allows us to rely on the Central Limit Theorem.
- Variance was addressed by using the default robust t-test in R (Welch's t-test).

1.4 Construct 95% confidence intervals for the mean of two groups and interpret them

- We are 95% confident that the true population mean of FC for Piston engines lies within that interval **(29.61919, 30.62560)**.
- Similarly, the 95% CI for Turbofan is **(8.443816, 8.588554)**.

The two intervals do NOT overlap. This provides visual evidence that the mean Fuel Consumption differs significantly between the two engine types.

1.5 Check the association between Model and Sales Region in the whole sample using proper method and interpret your findings

To check the association between the 2 categories we used the chi square as the two types of variable needed it. The code is shown below.

```
table_model_region <- table(air_data$Model, air_data$SalesRegion)
chisq_result <- chisq.test(table_model_region)
```

The obtained p-value > 0.05 ($\sim$0.73), so we fail to reject the null hypothesis and conclude that there is no statistically significant association between Model and Sales Region. The airplane models are distributed similarly across sales regions.

1.6 Filter your data only considering $BombardierCRJ200$ and $Cessna172$ model airplanes

Here we applied a simple filter and removed the empty buckets. this way we obtained a clean dataset with only the 2 models.

```
filtered_air_data <- air_data |>
  filter(Model %in% c("$Bombardier CRJ200$", "$Cessna 172$")) |>
  droplevels()
```

1.7 Check the distribution of Price across two categories of Engine type (You can apply transformation if you think it is required)

Here we applied a transformation to the price (log) since it appeared to be completely left-skewed. visually we can clearly conclude that there is a significant difference of Price between the two types of Engines. Data can be visualized in figures 6 and 7

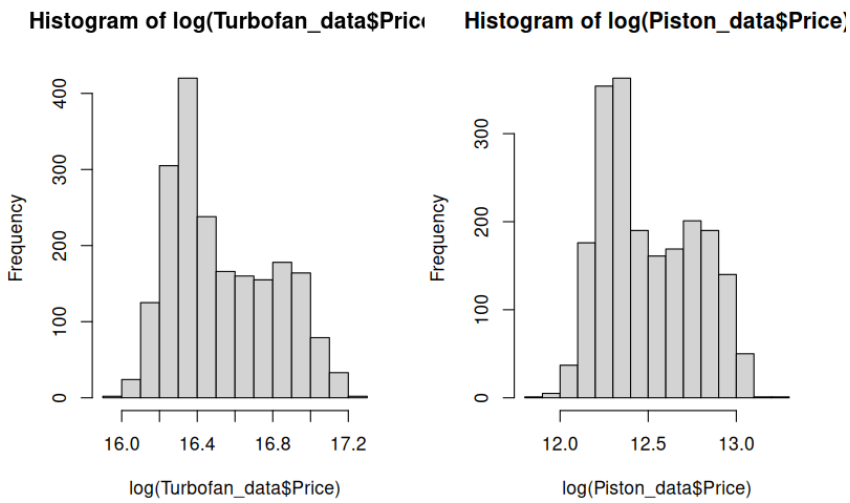

Figure 6: Histogram of Log Price by Engine Type

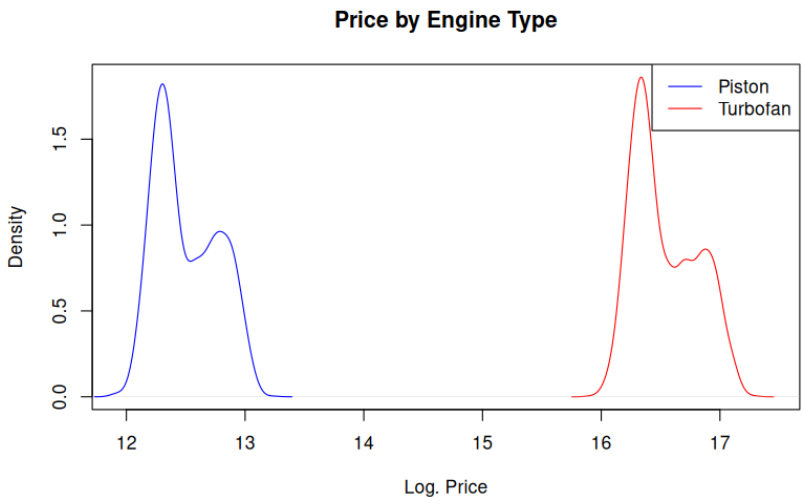

Figure 7: Distribution plot of Log Price by Engine Type

1.8 Categorize the variable Price into two categories as "low" and "High" by cutting from the median and save it as a new variable into your data frame

To compute the categorization of price we first calculated the median and then applied it to a new variable: $Price_Category$.

```
median_price <- median(filtered_air_data$Price, na.rm = TRUE)
filtered_air_data$Price_Category <- factor(
  ifelse(filtered_air_data$Price > median_price, "High", "Low")
)
```

1.9 Cross classify model and price categories and interpret the conditional probabilities

Row proportions P(PriceCategory | Model) table shows (table 4), for each model, what proportion of its airplanes fall into "Low" vs "High" price. Since $BombardierCRJ200$ (Turbofan) has much higher prices than $Cessna172$ (Piston), we expect nearly all $BombardierCRJ200$ to be in the "High" category and nearly all $Cessna172$ to be in the "Low" category. Nevertheless, using the median as a separator, some $BombardierCRJ200$ still fell into the "low" category because the number of the first model is slightly higher than that of the second model..

	High	Low
$BombardierCRJ200$	0.997074598	0.002925402
$Cessna172$	0.000000000	1.000000000

Table 4: Conditional probabilities: P(PriceCategory | Model) - row proportions

Column proportions P(Model | PriceCategory) table shows (table 5), for each price category, what proportion belongs to each model. This table confirms further the hypothesis of association between the model and price. The "low" category is dominated by $Cessna172$ and the "High" category by $BombardierCRJ200$, so it confirms a strong association between airplane model and price

	High	Low
$BombardierCRJ200$	1.000000000	0.002933985
$Cessna172$	0.000000000	0.997066015

Table 5: Conditional probabilities: P(Model | PriceCategory) - column proportions

1.10 Is there an association between the model of the airplane and its price level. Analyze it by using proper statistical method

we applied the chi square test and obtained a p-value as a result.

```
test_model_price <- chisq.test(cross_model_price)
```

The p-value < 0.05, so we reject the null hypothesis and conclude that there is a statistically significant association between airplane model and price level. Given that $BombardierCRJ200$ (Turbofan) costs millions while $Cessna172$ (Piston) costs hundreds of thousands, we expect a very small p-value, confirming a strong association.

1.11 Cross classify the variables Model and Sales region. Interpret the conditional probabilities and then test whether there is an association between these two characteristics

P(SalesRegion | Model) (row proportions, table 6) shows the distribution of sales regions for each model. Both models have similar proportions across regions, so we conclude that the sales region is independent of the model.

	Africa	Asia	Australia	Europe	North America	South America
$BombardierCRJ200$	0.1843003	0.1628474	0.1604096	0.1589469	0.1638225	0.1696733
$Cessna172$	0.1652771	0.1726336	0.1696910	0.1765571	0.1574301	0.1584110

Table 6: Conditional probabilities: P(SalesRegion | Model)

P(Model | SalesRegion) (column proportions, table 7) shows, within each region, what share belongs to each model. Both models appear in roughly equal proportions within every region, so this supports our independence assumption.

	Africa	Asia	Australia	Europe	North America	South America
$BombardierCRJ200$	0.5286713	0.4868805	0.4874074	0.4752187	0.5114155	0.5186289
$Cessna172$	0.4713287	0.5131195	0.5125926	0.5247813	0.4885845	0.4813711

Table 7: Conditional probabilities: P(Model | SalesRegion)

We applied a chi square test to further test the hypothesis. As expected, p-value > 0.05 ($\sim$0.29), so we fail to reject the null hypothesis and conclude there is no significant association. The two models are sold in similar proportions across regions.

2 Questions 2

2.1 Create a new data frame only including "Airbus $A320$", "Airbus $A350$", "Boeing 737" and "Boeing 777" models of airplanes. Check the distribution of Price: first for the observations in this sample and then for each model in the data frame. Interpret your findings.

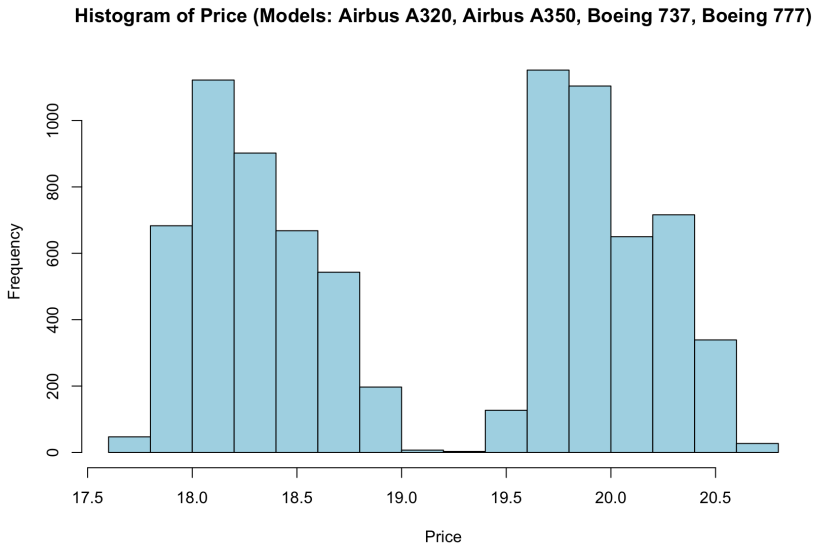

Figure 8: Price for all selected models

After the log transformation, Price is approximately normally distributed overall. Per model, Boeing 777 and Airbus $A350$ have higher mean log(Price) values reflecting their larger capacity and longer range. Boeing 737 and Airbus $A320$ cluster at lower price levels. Within each model, the log(Price) distribution is roughly symmetric, confirming that log transformation was appropriate for subsubsequent ANOVA analysis.

2.2 Analyze the numerical variables that are affected by the "Model". Test the assumptions of the statistical method, for the cases that you have found a significant association, by using corresponding tests and plots. Write your conclusions.

We first used a visual approach to filter the variables that had no significance. The boxplots in figure 10 show that the four models fall into two subsub-groups: cheaper and more expensive models. This difference created a change in some variables, but left untouched others. The next subsection highlights our analysis.

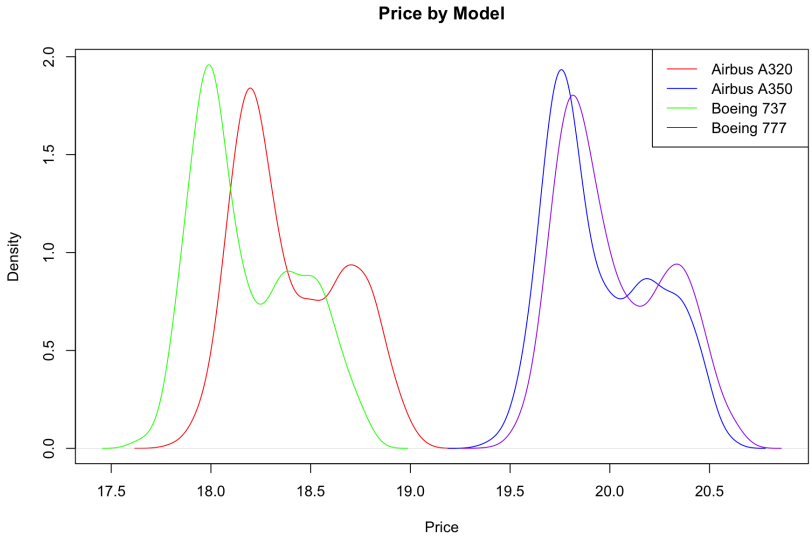

Figure 9: Price by model

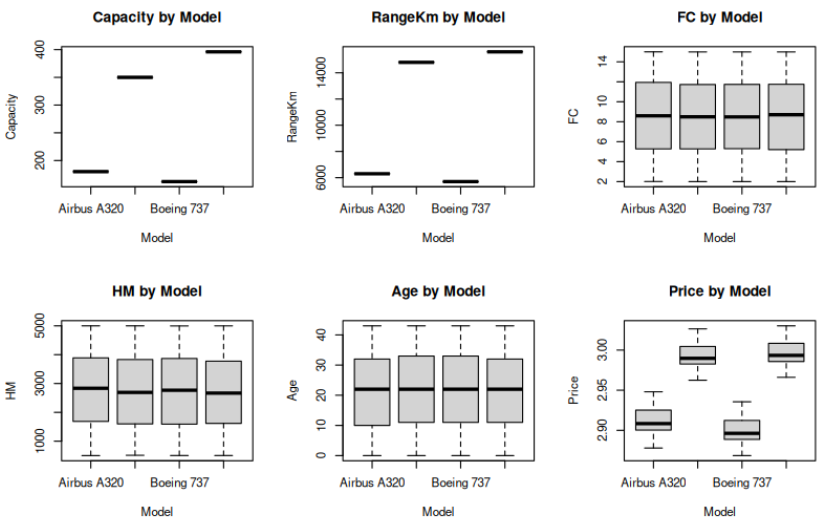

Figure 10: Boxplot of numerical variables

Variables Affected by Model

- **Price**: The $A350$ and $B777$ (expensive subsub-group) are in a different wealth bracket compared to the $A320$ and $B737$ (cheaper subsub-group).
- **Capacity and RangeKm**: These are affected by the model, though there is almost zero variation within groups in the dataset. This means Model, Capacity, and RangeKm are all telling the model the same thing (high multicollinearity).
- **FC, HM, and Age**: NOT significantly different across models since all four are Turbofan jets and the dataset appears to have similar distributions for these.

Assumption Analysis

We analyzed only the variable that passed the visula test.

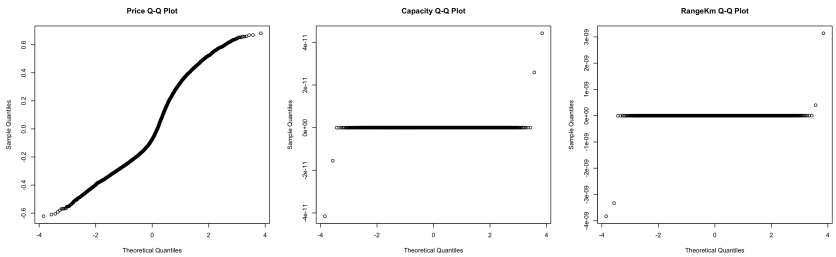

Figure 11: QQ-plot for all numerical variables

- The appearance of a "horizontal flat line" in the QQ-plots for **Capacity** and **RangeKm** occurs because these are fixed specifications for a given airplane model. Consequently, there is near-zero variance within each model group.
- For the case of **Price** the QQ-plot looks a bit curved with respect to the line meaning it might not be normally distributed. However, both Independence test and Homoscedasticity test show a p-value > 0.05 so we are assuming Normality, Independence and Homoscedasticity.

Conclusion

The ANOVA p-value for Price is < 0.05, so at least one model has a significantly different mean log(Price). The Tukey HSD test shows which specific pairs differ. We expect $Boeing777$ and $AirbusA350$ (long-range) to have significantly higher prices than $Boeing737$ and $AirbusA320$ (medium-range).

2.3 Apply a two-way ANOVA including Sales Region to the model. Interpret your findings.

We applied ANOVA and the results are reported in table 8. There is no significant interaction since the p-value is much higher than 0.05, so we can conclude that the relationship between the aircraft Model and its Price does not change based on the Sales Region. Also, while there might be a tiny hint of a regional difference, statistically speaking, Region does not significantly affect Price.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Model	3	6032	2010.7	28038.546	< 2e-16 ***
SalesRegion	5	1	0.2	2.142	0.0575 .
Model:SalesRegion	15	1	0.1	0.830	0.6445
Residuals	8263	593	0.1		

Table 8: ANOVA for Price and Region

2.4 Convert the variable Production Year to a categorical variable with two levels as "Older" and "Newer" and save it as a new variable named "year_cat" in the data frame.

To compute the categorization of Production Year we first calculated the median and then applied it to a new variable: *year_cat*.

```
median = median(air_data$ProductionYear)
air_data$year_cat <- as.factor(
    ifelse(air_data$ProductionYear < median, "Older", "Newer")
)
```

2.5 Analyze the effect of Model and year ("year_cat") together on the price. Analyze whether the interaction of two term is significant. Interpret your findings.

Figure 12 plot aligns with previous analysis: there are two cheaper plane models and two expensive ones, and now we observe that there are older and newer models also which are a bit more expensive (shifted to the right) in each case. The ANOVA interaction plot shown in figure 13 is just one more analysis that make us sure about what has been deducted before.

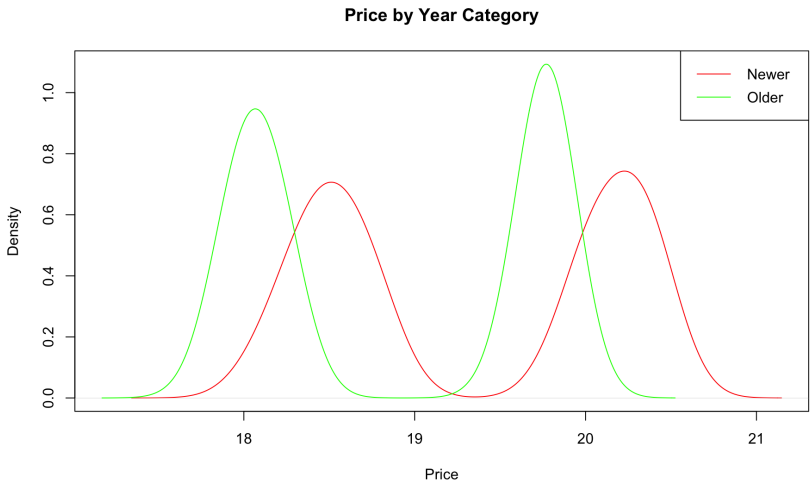

Figure 12: Plot by year_cat

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Model	3	6032	2010.7	78235.015	< 2e-16 ***
year_cat	1	381	381.4	14838.647	< 2e-16 ***
Model:year_cat	3	0	0.0	0.984	0.399
Residuals	8279	213	0.0		

Table 9: ANOVA for Price by Model and Year

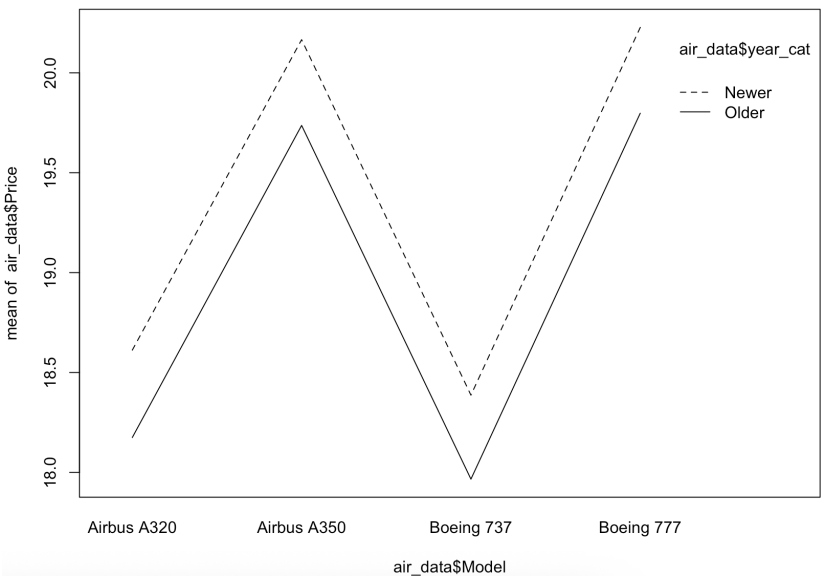

Figure 13: ANOVA Interaction Plot

Conclusion

The two-way ANOVA with interaction tests three effects:

Based on the two-way ANOVA results, both the airplane Model and the categorized production year (`year_cat`) have a significant independent effect on the Price. However, the interaction between the Model and the production year is not statistically significant, indicating that the price difference between newer and older airplanes remains consistent regardless of the specific airplane model.

Post-hoc tests

All three post-hoc tests (Tukey, Bonferroni, and LSD) confirm significant price differences between the "cheaper" models ($A320$, $B737$) and the "expensive" models ($A350$, $B777$). However, there is likely no significant price difference *between* the two cheaper models themselves, or *between* the two expensive models. The results are consistent across the different penalty methods.

3 Questions 3

3.1 Consider the numerical variables in the data set and find the best SIMPLE linear regression model to predict the prices. Test the assumptions and use transformations if it is required. Explain why the model you find is the best simple linear regression model and interpret the model coefficients.

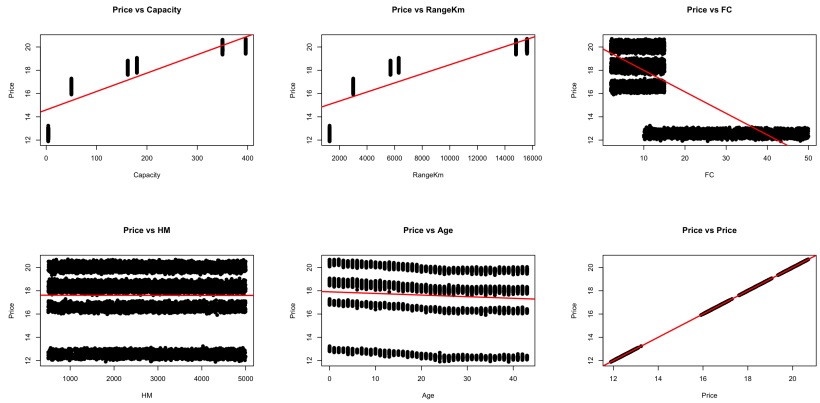

Figure 14: Price plots analysis

According to plot analysis (figure 14), we selected the variables `Capacity`, `RangeKm`, and `FC` (Fuel Consumption) as the best candidates for the linear model because they show the most linear relationship with `Price`. A log() transformation was applied to `Price` to normalize its distribution.

The correlation matrix confirmed that `Capacity`, `RangeKm`, and `FuelConsumption` are the top 3 numerical variables with the highest correlation to `Price`.

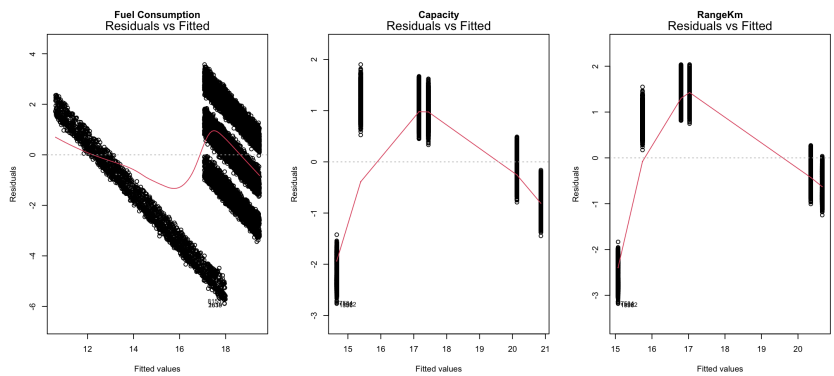

Figure 15: Residual vs Fitted plot

The Residual vs Fitted plots (figure 15) suggested that the model may need a quadratic term.

After comparing `Capacity` vs `RangeKm` using a quadratic term (figure 16), the ANOVA shows that `RangeKm` + `RangeKm`^{2} yields the lowest RSS results.

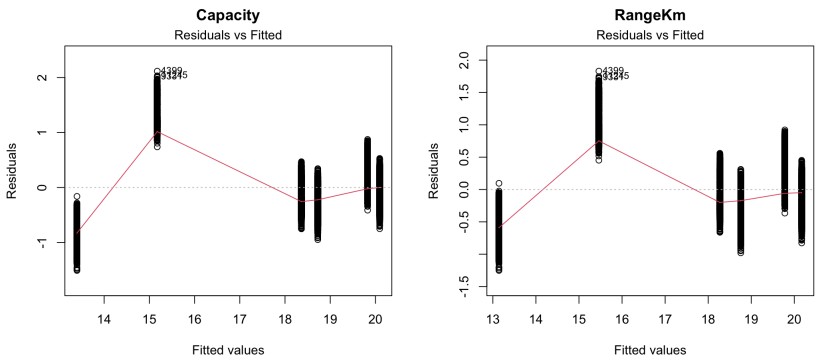

Figure 16: Residual vs Fitted plot (Quadratic term)

The new model with the quadratic term shows better residuals.

Regression assumptions analysis

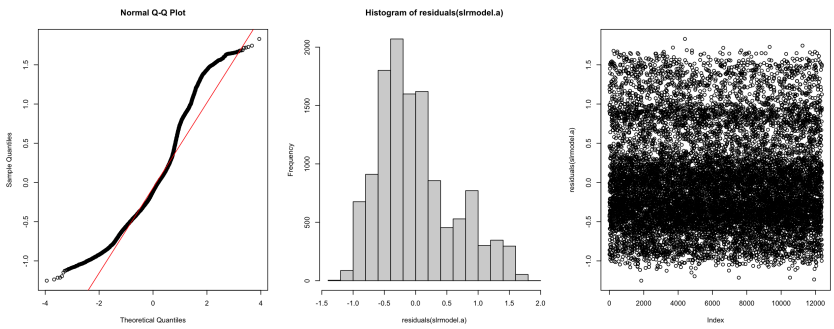

Figure 17: Regression assumptions analysis

This model shows p-value < 0.05 on the Breusch-Pagan test, meaning it has signs of heteroscedasticity.

Conclusion

We selected `RangeKm` (along with its quadratic transformation) as the best single predictor because it shows a high correlation with Price and explains the highest proportion of the variance in airplane prices. Results reported in table 10.

- **Intercept (β_0)**: The expected log(Price) or baseline price of a plane.
- **Slope for RangeKm (β_1 & β_2)**: For each additional `RangeKm`, the log(Price) increases by β_1 and β_2, meaning Price increases exponentially by e^{β_2}.

Residuals				
Min	1Q	Median	3Q	Max
-1.2519	-0.4347	-0.1157	0.2976	1.8296
Coefficients	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	1.113e+01	1.752e-02	635.5	< 2e-16 ***
RangeKm	1.654e-03	5.508e-06	300.4	< 2e-16 ***
RangeKm_2	-7.052e-08	3.023e-10	-233.3	< 2e-16 ***
Model Statistics				
Residual standard error: 0.6079 on 12374 degrees of freedom				
Multiple R^2: 0.944, Adjusted R^2: 0.944				
F-statistic: 1.043e+05 on 2 and 12374 DF, p-value: < 2.2e-16				

Table 10: Summary of Linear Regression (Model A with Quadratic Term)

3.2 Fit a multivariate linear regression model with the most important two (numerical) variables. Use transformations if it is needed and test all the assumptions. Then compare this model to the simple linear regression model that you fit in (a). Which one is a better model? Why?

First, we checked a model against all our top variables: `Capacity`, `RangeKm`, `FC`, and `Age`, table 11.

```
regmodel.all <- lm(Price ~ ., data = numeric_air_data)
```

Residuals				
Min	1Q	Median	3Q	Max
-2.8521	-0.3464	0.1172	0.5222	1.6157
Coefficients	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	1.682e+01	2.569e-02	654.989	< 2e-16 ***
Capacity	3.080e-02	3.505e-04	87.856	< 2e-16 ***
RangeKm	-4.671e-04	8.884e-06	-52.576	< 2e-16 ***
FC	-8.760e-02	8.055e-04	-108.751	< 2e-16 ***
HM	2.249e-06	5.190e-06	0.433	0.665
Age	-1.819e-02	5.258e-04	-34.590	< 2e-16 ***
Model Statistics				
Residual standard error: 0.7465 on 12371 degrees of freedom				
Multiple R^2: 0.9156, Adjusted R^2: 0.9155				
F-statistic: 2.683e+04 on 5 and 12371 DF, p-value: < 2.2e-16				

Table 11: Summary of Multiple Linear Regression (All Predictors)

```
regmodel.best4 <- lm(Price ~ Capacity + RangeKm + FC + Age, data = numeric_air_data)
```

```
anova(regmodel.all, regmodel.best4)
```

The ANOVA confirmed that the model with all predictors is not significantly better than the

model with these 4 predictors. However, checking Variance Inflation Factors (VIF) revealed critical multicollinearity (tables 13 and 12).

```
cor(numeric_air_data[,-price.col])
vif(regmodel.best4)
```

Table 12: Correlation Matrix for Numeric Variables

	Capacity	**RangeKm**	**FC**	**HM**	**Age**
Capacity	1.000	0.990	-0.468	-0.012	0.019
RangeKm	0.990	1.000	-0.424	-0.014	0.019
FC	-0.468	-0.424	1.000	-0.011	-0.020
HM	-0.012	-0.014	-0.011	1.000	0.010
Age	0.019	0.019	-0.020	0.010	1.000

Table 13: Variance Inflation Factors for Best 4 Model

Predictor Variable	**VIF**
Capacity	55.540733
RangeKm	52.879523
FC	1.413722
Age	1.000534

The Correlation Matrix and VIF tell us that Capacity and RangeKm are highly correlated. VIF also tell us that there is a multicollinearity problem within our predictors. We choose to drop the predictor with the highest VIF: `Capacity`.

Now, VIF shows no multicollinearity among the selected predictors.

By systematically dropping predictors and comparing via ANOVA, we discovered that keeping `RangeKm` and `FC` led to higher R^2 and a smaller Sum of Square Residuals. Similar to the SLR, adding a quadratic term `RangeKm`^{2} further improved the model.

Model A vs Model B comparison

Model 1: Price $\sim$ RangeKm + RangeKm_2
Model 2: Price $\sim$ RangeKm_2 + RangeKm + FC

	Res.Df	**RSS**	**Df**	**Sum of Sq**	**F**	**Pr(>F)**
1	12374	4572.8				
2	12373	3321.5	1	1251.3	4661.3	< 2.2e-16 ***

Table 14: Analysis of Variance Table: Model A vs Model B

- The MLR model with `RangeKm` + `RangeKm`^{2} + `FC` has a higher adjusted R^2 and a lower residual standard error than the SLR with Capacity alone, indicating a better fit.

- The ANOVA F-test confirms that adding the second variable significantly improves the model.

The assumptions can be visualized in figure 18.

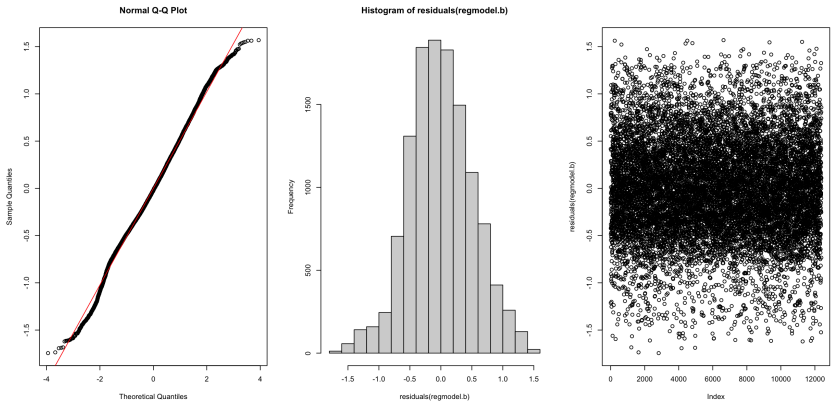

Figure 18: Assumptions analysis for Model B

3.3 Encode the variable model into three categories considering whether the plane is "Airbus", "Boeing" or "Other". Now add this model factor to the regression model you have chosen in subsection (b). Interpret the coefficients and overall summary of the model. Compare the model in subsection (b) with the model that has an additional factor. Which one would you choose? Why?

We added `ModelCat` to our selected model. However, introducing `ModelCat` made `FC` statistically insignificant. We dropped `FC` to simplify the model, yielding: `lm(Price ~ RangeKm_2 * ModelCat + RangeKm)`. lm result is reported in table 15, while the residuals are in figure 19.

Interpretation

- **Intercept** (β_0): Expected log(Price) for the baseline category.
- **ModelCat effects**: The coefficients for Boeing and "Other" represent the shift in the baseline log(Price) compared to Airbus.
- **Interaction** (`RangeKm_2`$\times$`ModelCat`): The slope for the squared range changes depending on the airplane manufacturer.

The Residual vs Fitted plot shows good results.

Residuals				
Min	1Q	Median	3Q	Max
-0.62176	-0.21716	-0.07364	0.22958	0.75755
Coefficients	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	5.233e+00	2.792e-02	187.46	< 2e-16 ***
RangeKm_2	-1.288e-07	2.970e-10	-433.75	< 2e-16 ***
ModelCatBoeing	5.520e-01	1.009e-02	54.69	< 2e-16 ***
ModelCatOther	3.755e+00	1.741e-02	215.72	< 2e-16 ***
RangeKm	2.901e-03	6.150e-06	471.76	< 2e-16 ***
RangeKm_2:ModelCatBoeing	1.332e-09	6.122e-11	21.76	< 2e-16 ***
RangeKm_2:ModelCatOther	NA	NA	NA	NA
Model Statistics				
Residual standard error: 0.2686 on 12371 degrees of freedom				
Multiple R^2: 0.9891, Adjusted R^2: 0.9891				
F-statistic: 2.24e+05 on 5 and 12371 DF, p-value: < 2.2e-16				

Table 15: Summary of Multiple Linear Regression (Model C Interaction)

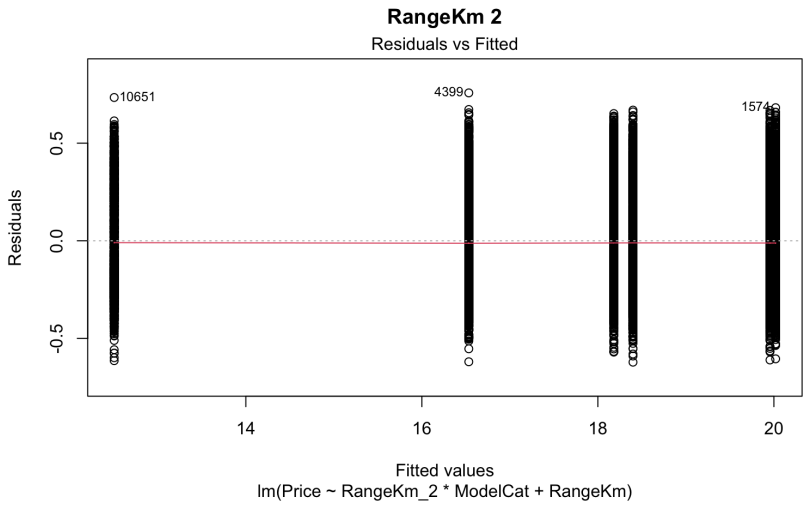

Figure 19: Residual vs Fitted for Model C

Model B vs Model C comparison

Model 1: Price ∼ RangeKm_2 + RangeKm + FC						
Model 2: Price ∼ RangeKm_2 * ModelCat + RangeKm + FC						
	Res.Df	**RSS**	**Df**	**Sum of Sq**	**F**	**Pr(>F)**
1	12373	3321.5				
2	12370	892.2	3	2429.3	11227	< 2.2e-16 ***

Table 16: Analysis of Variance Table: Model B vs Model C

- The ANOVA analysis tells us that adding `ModelCat` decreases the Residual Sum of Squares, meaning the model significantly improves with this new variable.
- We selected this interaction model (Model C) as our best model.

3.4 Test the validity of the final model that you choose.

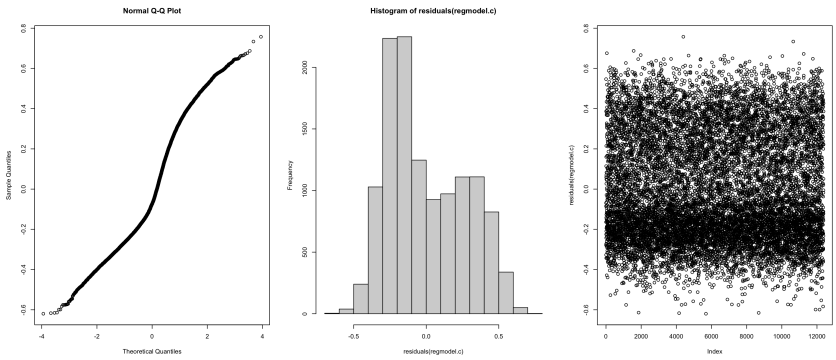

Figure 20: Validity test for final model C

To fully validate the predictive power of our chosen model (`Price ~ RangeKm_2 * ModelCat + RangeKm`), we split the data into an 80% training set and a 20% testing set.

Validity and Performance

- **Out-of-sample MSE**: We calculated the Mean Squared Error against the holdout Test Set, which quantifies the average squared prediction error on unseen data.
- **Diagnostics**: The Q-Q plot of the final model mostly follows the diagonal (normality holds). Durbin-Watson statistic is close to 2 (no autocorrelation).
- Plotting `Actual vs Predicted` (figure 21) values from the test set showed data points clustering closely around the ideal $y = x$ reference line, meaning the model generalizes well and confirms our findings.

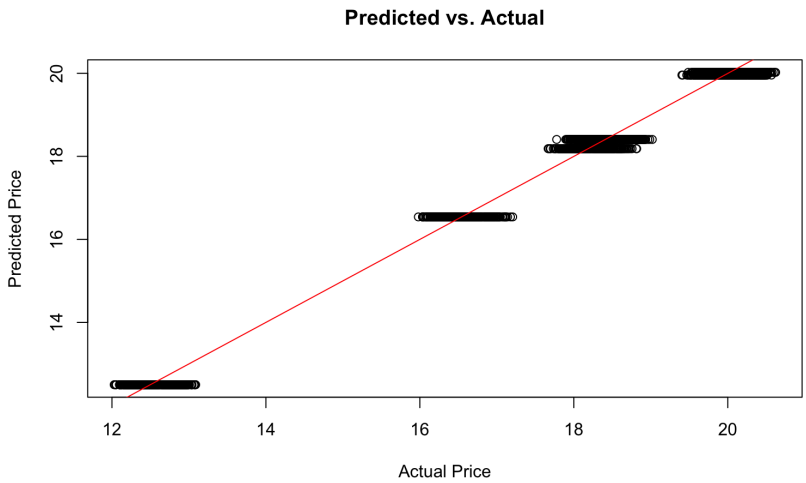

Figure 21: Actual vs Predicted values

4 Questions 4

4.1 Apply PCA analysis on airplane data and interpret the results of the analysis.

We applied a log transformation to the Price variable, consistent with the methodology in Q3, even though it was not utilized directly as an input for the PCA.

To ensure a robust analysis, we performed the PCA using two functions: `princomp()` and `PCA()`. Both methods yielded identical results, identifying five principal components based on the calculated eigenvalues (table 17).

	Dim.1	**Dim.2**	**Dim.3**	**Dim.4**	**Dim.5**
Variance	2.298	1.011	0.989	0.693	0.009
% of var.	45.952	20.226	19.779	13.858	0.185
Cumulative % of var.	45.952	66.177	85.957	99.815	100.000

Table 17: Eigenvalues

Discussion of Variance and Selection Criteria

As shown in table 17, Principal Component 1 (PC1) accounts for 45.95% of the total variance, while PC2 and PC3 explain approximately 20% each. To determine the optimal number of components to retain, we evaluated three standard criteria:

- **80% Variance Rule**: To capture at least 80% of the information, we must retain the first three components, which collectively explain 85.96% of the total variation.

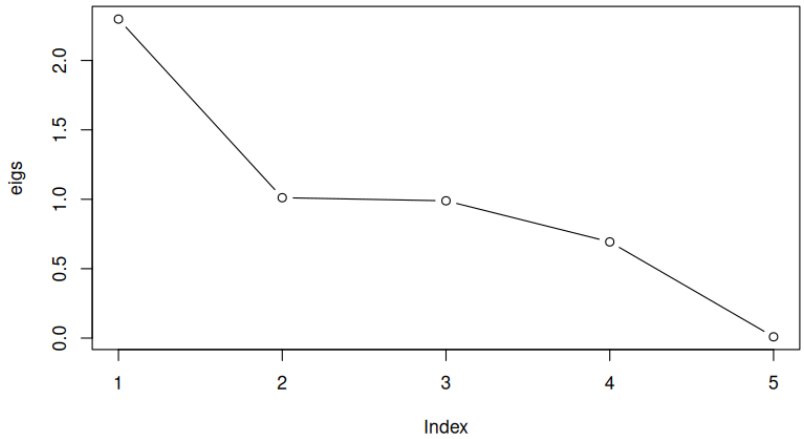

Figure 22: Scree plot of eigenvalues

- **Kaiser's Rule**: While PC1 and PC2 strictly meet this requirement, PC3 is a borderline case with an eigenvalue of 0.989 (see table 17). Given how close this value is to 1, retaining PC3 is statistically justifiable to meet our variance threshold.
- **Elbow Rule**: If we were to strictly follow the "elbow" or bend in the scree plot (see Figure 22), the most significant drop occurs after PC1.

Balancing these methods, we would retain the first three components to ensure a comprehensive representation of the dataset's variance.

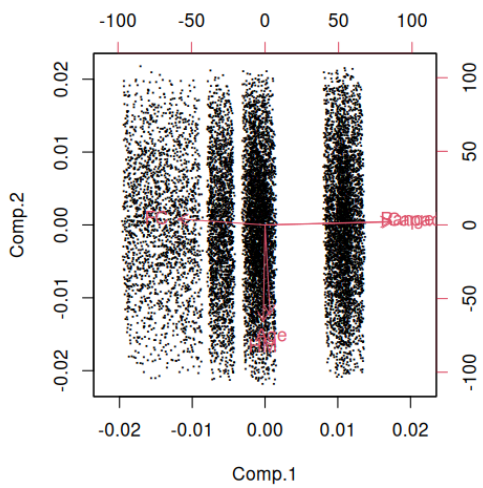

Figure 23: PCA Biplot with individual observations

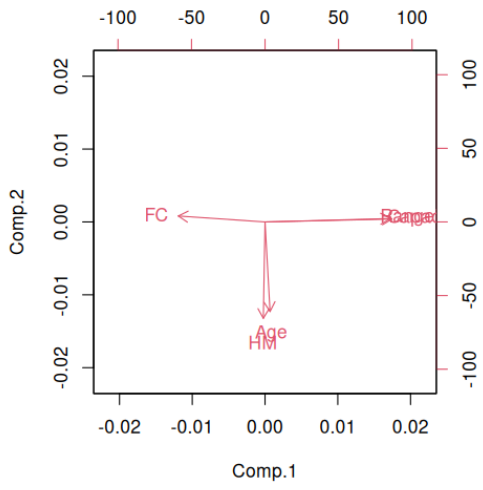

Figure 24: PCA Variables Factor Map

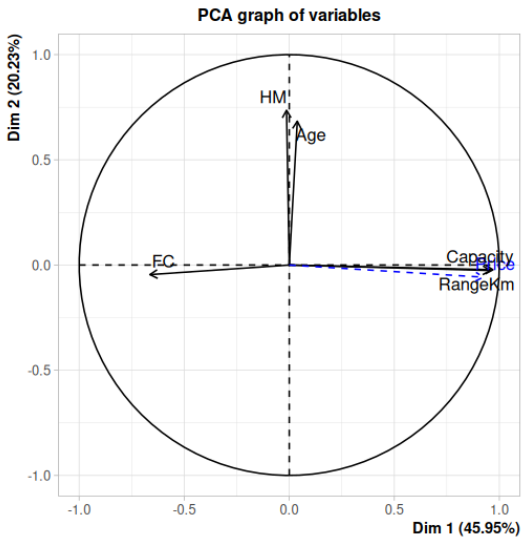

Figure 25: PCA Correlation Circle with Price as a supplementary variable

Interpretation of components

Figure 23 provides a biplot that includes the individual observations. We can observe that the data points are spread across the first two components, clearly forming four distinct clusters.

In Figure 25, we included Price as a supplementary variable (indicated by the blue dashed line). While Price was not used to calculate the principal components to avoid biasing the model, it is highly correlated with the first dimension. Specifically, it aligns closely with Capacity and RangeKm, confirming that higher-capacity, longer-range vehicles tend to have a higher price point, while being inversely related to FuelConsumption (FC).

The first dimension captures the scale and performance of the aircraft, and seems to be the

primary driver of Price. While the second dimension could represent the operational lifecycle of the airplane. Because the second dimension is orthogonal to the price, price is not affected by the operational lifecycle, age and Hourly Maintenance (HM).

Interpretation of quadrants

- **Top-Right**: Large, high-range planes that are older and expensive to maintain.
- **Bottom-Right**: Large, modern, high-range planes with relatively lower maintenance needs.
- **Top-Left**: Small, fuel-efficient planes that are older and need a lot of work.
- **Bottom-Left**: Small, modern, highly efficient planes (the entry-level newer models).

Assumptions

Variable	**MSA**
Capacity	0.518
RangeKm	0.520
FC	0.633
HM	0.563
Age	0.825
Overall KMO	**0.533**

Table 18: Kaiser-Meyer-Olkin (KMO) Factor Adequacy

Statistic	**Value**
Chi-Squared (χ^2)	52174.84
Degrees of Freedom (df)	10
p-value	0

Table 19: Bartlett's Test of Sphericity

- **KMO Test** (Figure 18): The overall KMO value is 0.533, which is below 0.6. Notably, Age shows the highest individual adequacy at 0.824.
- **Bartlett's Test** (Figure 19): Our result shows a p-value of 0 ($p < 0.05$), confirming that the variables are sufficiently related to justify PCA.
- **Linearity** (Figure 26): PCA assumes linear relationships between variables. The scatterplot matrix confirms a strong linear trend between Capacity and RangeKm, though relationships with other variables are less defined.

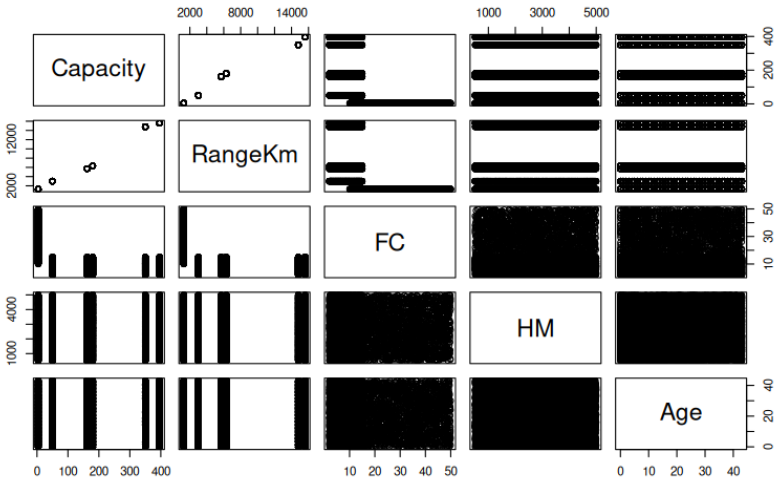

Figure 26: Linearity check via scatterplot matrix

4.2 Find the best linear model to predict price on the principal components. Do not forget to test the assumptions and the validity of the model.

Coefficients	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	18.238469	0.009318	1957.36	< 2e-16 ***
Comp.1	1.492014	0.004472	333.60	< 2e-16 ***
Comp.1_2	-0.273755	0.002826	-96.87	< 2e-16 ***

Model Statistics
Residual standard error: 0.7435 on 12374 degrees of freedom
Multiple R^2: 0.9162, Adjusted R^2: 0.9162
F-statistic: 6.767e+04 on 2 and 12374 DF, p-value: < 2.2e-16

Table 20: PCR Model Summary (Quadratic)

We evaluated several Principal Component Regression (PCR) specifications, concluding that the quadratic model (Figure 20) offers the most balanced performance.

Initially, we fitted a simple linear regression using only the first principal component (PC1), which captures the primary variance in Price (as shown in the Correlation Circle, Figure 25). While initial results were promising, the Residuals vs. Fitted plot revealed a distinct parabolic trend, signaling a violation of the linearity assumption. To address this, we introduced a quadratic term ($PC1^2$), which resulted in a substantial improvement in model fit (Figure 27). Both ANOVA (Partial F-test) and AIC comparisons statistically confirmed that the inclusion of the squared term was necessary to capture the non-linear relationship.

Regarding the remaining components, we opted to exclude PC2 and PC3. PC2 is orthogonal to Price, offering no predictive power, while PC3 was primarily defined by Age and Hourly

Maintenance (HM), variables which, in this specific dataset, showed weak correlation with the target. Even adding PC4 (mainly representing Fuel Consumption) showed some improvement in AIC, we ultimately chose to exclude it to prioritize model simplicity. By maintaining a simpler model, we reduce the risk of overfitting and ensure that the regression remains interpretable and generalizes effectively to new, out-of-sample data. PC5 was not utilized because of its low explanation in variance.

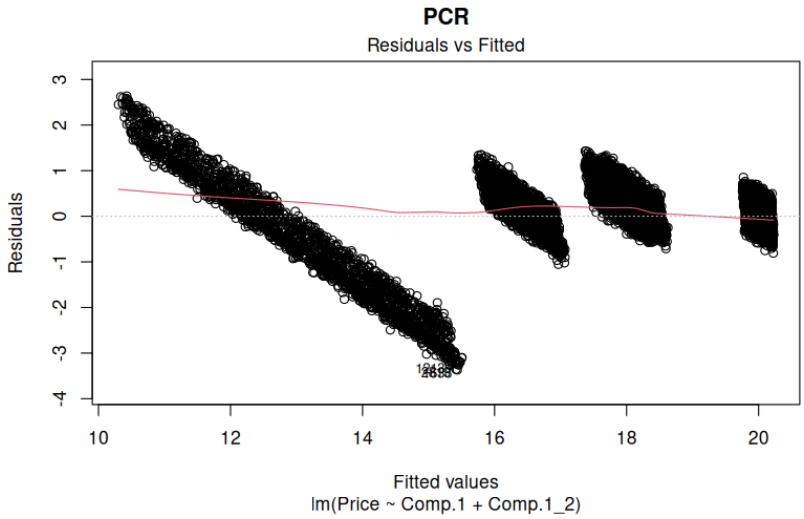

Figure 27: PCR Residuals Analysis

We don't need to test collinearity with `vif()` because components are orthogonal between them (no correlation) and there is no linear association between a variable and its squared term.

Assumptions

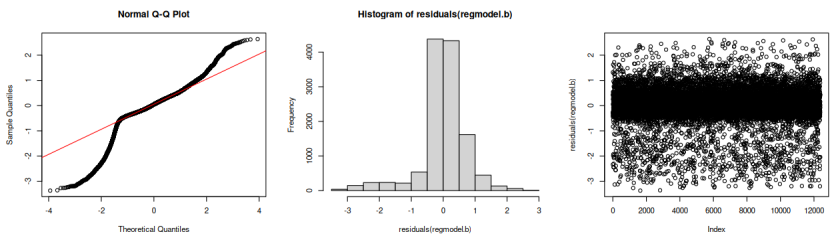

Figure 28: PCR Assumptions Diagnostics

- **Normality** (Figure 28 first and second plot): The histogram and Q-Q plot indicate that the residuals are generally normal but moderately left-skewed. While not a perfect bell curve, the large sample size invokes the Central Limit Theorem.
- **Homoscedasticity** (Figure 28 third plot): Although the Breusch-Pagan test suggests

heteroscedasticity due to its extreme sensitivity to large samples, the Residuals vs Index plot shows a consistent spread across the entire range. We conclude the variance is sufficiently constant.

- **Independence of errors**: The Durbin-Watson test yielded a p-value of 0.4675, failing to reject the null hypothesis of no autocorrelation.

Validity

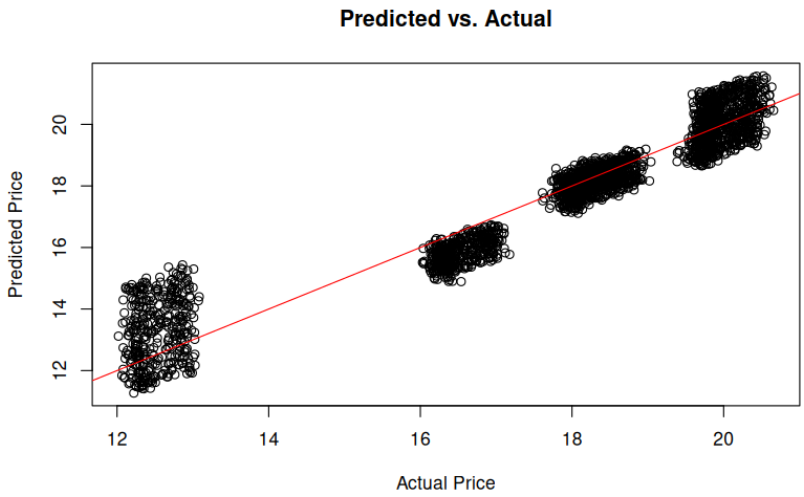

Figure 29: PCR Actual vs Predicted Prices

We performed a validation using an 80/20 train-test split. Figure 29 displays the relationship between the Actual Prices in the test set and the Prices Predicted by the model.

The model achieved a Mean Squared Error (MSE) of 0.53. While there is visible dispersion, this is a respectable result considering the model's simplicity. It relies on only two predictors (the first principal component and its quadratic term) to explain the majority of the price variation.

As seen in the plot, the observations are clustered into four distinct groups, maybe because of airplanes models.

4.3 Would you prefer the linear model that you fit in the final step of question 3 or this one? Explain why.

We prefer the Multiple Linear Regression (MLR) model from Question 3. While the Principal Component Regression (PCR) model is efficient using only two predictors ($PC1$ and $PC1^2$) to explain the data, the MLR model provides a much higher level of precision.

The performance gap is most evident in the Mean Squared Error (MSE). The MLR model achieved an MSE of 0.071, whereas the PCR model produced a significantly higher error of 0.527.

Although the MLR model introduces more complexity by using six variables (including the interaction between Model and RangeKm) and is very probably overfitting, in this case, we prioritize predictive accuracy over simplicity of the PCR model.

The only case we will prefer the PCR model is in case we have to predict Prices of airplanes from out-of-sample data and the airplane does not belong to a model in the trained dataset.

Conclusion

In this assignment, we analyzed the variables that affect airplane prices. We found a strong association between Price and variables like Model and RangeKm. We also confirmed that Engine Type significantly affects Fuel Consumption, while Sales Region has no significant effect on Price.

To predict airplane prices, we evaluated two main approaches. First, we built a Multiple Linear Regression (MLR) model using RangeKm, its quadratic term, and Model categories. Second, we applied Principal Component Analysis (PCA) and fitted a Principal Component Regression (PCR) model using the first principal component.

The MLR model achieved a Mean Squared Error (MSE) of 0.071 on the test set. The PCR model produced a higher MSE of 0.527. While the PCR model is simpler and uses fewer predictors, we prioritize predictive accuracy. Therefore, we selected the MLR model as the best approach to predict airplane prices for this dataset.

A Appendix section

Use of AI

- Gemini (Pro): sentence enhancing, crosschecking personal assumptions